

IMO SSE 12 Meeting Summary

26 March 2026



The 12th session of the International Maritime Organization Sub-Committee on Ship Systems and Equipment (SSE 12) was held 9 to 13 March 2026 at the IMO Headquarters in London, supplemented by hybrid (remote) arrangements.

LISCR participated in the following groups, in addition to the plenary.

ID	Subject
WG 1	Working Group on Life-Saving Appliances
WG 2	Working Group on Fire Protection
WG 3	Working Group on Development of a GHG Safety Framework

Opening

Attacks on ships in the vicinity of the Strait of Hormuz

Several delegations expressed views on the ongoing conflicts and their negative impact on international shipping and seafarers. Many delegations raised concerns regarding the safety of ships and crews following attacks on commercial vessels in the Strait of Hormuz and nearby waters and commended the Secretary-General for drawing international attention to the situation.

Delegations also emphasized the importance of respecting international law and freedom of navigation and highlighted the severe human consequences of the escalation, stressing that seafarers, as key workers, should not be adversely affected by geopolitical conflicts.

Ventilation of partially enclosed lifeboats and liferafts

107th session of the Maritime Safety Committee (MSC 107) adopted amendments to the Life-Saving Appliances (LSA) Code (MSC.535(107)) to mandate ventilation of totally enclosed lifeboats and amendments to the *Revised recommendations on testing of lifesaving appliances* (MSC.81(70)). The amendments entered into force on 1 January 2026 and will be applicable to totally enclosed lifeboats installed on or after 1 January 2029. Issues associated with partially enclosed lifeboats were left for further discussion.

Amendments to the LSA Code

SSE 12 considered draft amendments to the LSA Code concerning ventilation requirements for partially enclosed lifeboats. The draft amendments would require partially enclosed lifeboats installed on or after 1 January 2032 to provide sufficient

ventilation to maintain CO₂ concentration below 5,000 ppm for the full complement of persons, either through powered ventilation capable of delivering at least 5 m³/h per person for at least 24 hours, ventilation openings, or other equivalent solutions. During the discussion, differing views were expressed on the appropriate basis for sizing ventilation openings, including the proposed “4% of floor area” or “0.015 m² per person”. SSE 12 agreed to remove the “4% of floor area” figure and retain “0.015 m² per person” only as an illustrative example pending supporting evidence and invited interested Member States and organizations to submit technical evidence to SSE 13 to determine the appropriate figure, failing which the prescriptive provision on ventilation openings may be deleted.

Amendments to the test requirements for type approval

SSE 12 considered draft amendments to the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)) for partially enclosed lifeboats. SSE 12 agreed in principle to incorporate the proposals and to further develop test procedures. These procedures are intended to verify that ventilation arrangements maintain CO₂ concentration below a time-weighted average of 5,000 ppm.

The draft test method includes environmental simulation conditions such as:

- Lifeboat closed and stationary;
- Wind conditions not exceeding Beaufort force 2;
- Simulated occupants using displacement forms; and
- Controlled CO₂ generation rates.

The use of heat sources to simulate body heat was considered optional.

Fixed carbon dioxide fire-extinguishing system

SSE 12 further considered the revision of the Revised guidelines for the maintenance and inspection of fixed carbon dioxide fire-extinguishing systems (MSC.1/Circ.1318/Rev.1) and agreed a number of technical clarifications to the draft text. These included:

- Replacing “head valve” with “cylinder valve”;
- Retaining the term “storage container” to ensure low-pressure systems remain within scope;
- Simplifying hose inspection requirements to checks for visible cracks and damage; and
- Requiring verification that markings remain clearly readable during annual inspections.

With regard to minimum recommended maintenance, SSE 12 did not agree to require 10-year maintenance to be conducted in dry dock and deleted the reference. It also replaced “and/or” with “and” to clarify that maintenance is to be carried out at both periodical and renewal surveys. In addition, SSE 12 agreed on a hydrostatic testing regime requiring testing at intervals not exceeding 10 years from the date of the previous test.

SSE 12 further agreed that the anniversary date for hydrostatic testing should be based on the manufacturing date of the cylinder rather than the installation date. It also clarified inspection and testing requirements following repair or replacement of components.

SSE 12 subsequently prepared the draft amendments to MSC.1/Circ.1318/Rev.1 on the Revised guidelines for the maintenance and inspection of fixed carbon dioxide fire-extinguishing systems, inviting Member States to apply the revised guidelines not earlier than 12 months after approval, with a view to approval by MSC 112.

Revision of SOLAS Chapter III and the LSA Code

SSE 12 continued work on the goal-based revision of SOLAS chapter III and the LSA Code to address gaps and inconsistencies in the current requirements. SSE 12 noted that a full gap analysis can only proceed once functional requirements (FRs) and expected performances (EPs) are finalized and therefore agreed to develop a provisional road map focused on completing this analysis. The work remains at an early stage, with an intersessional working group to be established and the target completion year provisionally extended to 2030.

Self-righting or canopied reversible liferaft

SOLAS requires liferafts on ro-ro passenger ships to be either automatically self-righting, or canopied reversible liferafts which can operate safely whichever way they are floating. Similar requirements are currently not applicable to other ship types. For larger conventional liferafts, righting in an emergency may be challenging due to injuries, exhaustion, adverse sea conditions or the dimensions. Following the instruction of the Maritime Safety Committee (MSC), the SSE Sub-Committee worked on this subject. SSE 11 (March 2025) agreed to require self-righting or canopied reversible liferafts for passenger and cargo ships for future installations, except for liferafts with a capacity up to 12 persons.

SOLAS Convention (chapter III)

SSE 12 developed draft amendments to SOLAS chapter III to require that liferafts on new ships be either automatically self-righting or canopied reversible liferafts. SSE 12 agreed that the application date should be expressed as “ships constructed on or after” rather than “liferafts installed on or after”, reflecting that the requirement applies only to new ships. At the same time, the wording “replaced on or after” was retained to ensure that liferafts replaced on existing ships would not need to comply with the new requirements. SSE 13 will continue to work on this.

LSA Code

SSE 12 developed draft amendments to chapter IV of the LSA Code, introducing additional technical requirements for automatically self-righting liferafts and canopied reversible liferafts, including provisions on stability, canopy arrangements, self-draining capability and accessibility of equipment regardless of the raft’s orientation. SSE 13 will continue to work on this.

2000 HSC Code

SSE 12 agreed that the new requirements for automatically self-righting and canopied reversible liferafts should also apply to high-speed craft under the 2000 HSC Code. However, as an application clause would be required to limit the provisions to new craft, the detailed drafting was deferred to the Correspondence Group under the current output.

Gasket

Paragraph 2.1.2.5 of chapter 5 of the FSS Code requires all discharge piping components, including gaskets, to withstand a temperature of 925°C. This approach is not consistent with other IMO instruments governing fixed fire-extinguishing systems, which expressly exclude gaskets from equivalent high-temperature performance requirements, for example, the IGC Code.

SSE 12 agreed to the draft amendment to paragraph 2.1.2.5 of chapter 5 of the FSS Code to exclude gaskets from the requirement to be made of materials having a melting temperature which exceeds 925°C, applicable to both new and existing ships, for approval by MSC 112 and subsequent adoption by MSC 113.

Fire protection system and materials

Plastic pipes

SSE 12 agreed draft amendments to the 2010 Fire Test Procedures Code (2010 FTP Code) to introduce fire test procedures for plastic pipes in annex 1, part 6. SSE 12 agreed that the FTP Code should reference the Guidelines for the use of plastic pipes on ships (resolution A.753(18)), as amended, to ensure the reference remains valid while the Sub-Committee on Ship Design and Construction (SDC) is revising those Guidelines. During the discussion, concerns were raised that the proposed fire testing requirements might be restrictive and could limit the types of plastic pipes able to pass the test. SSE 12 clarified that the FTP Code establishes fire test procedures only, while any mandatory requirements for the use of plastic pipes on board ships would need to be specified in SOLAS. SSE 12 agreed on the draft amendments introducing fire test procedures for plastic pipes for submission to MSC 112 for approval and subsequent adoption.

H class division

SSE 12 considered draft amendments to the 2010 Fire Test Procedures Code (2010 FTP Code) to introduce fire test procedures for “H” class divisions in annex 1, part 3, including consequential amendments and potential implications for SOLAS. SSE 12 agreed on several elements of the draft text, including the introduction of references to “H” class divisions in headings and application provisions, a provision that tests for “H” class divisions should follow the same requirements as “A” class divisions unless otherwise specified, and draft performance criteria. However, discussion on the definition of the hydrocarbon fire test and the appropriate fire testing curves could not be finalized due to time constraints, and SSE 12 tasked further consideration by the FP Correspondence Group¹.

Review and update SOLAS regulation II-2/9

SOLAS Regulation II-2/9 focuses on the containment of fire on ships, i.e. to ensure that a potential fire is contained within the space where it originated by providing requirements to fire integrity and insulation of boundaries, including openings and penetrations within those boundaries. Since the adoption of the current regulation in 2000, numerous Unified Interpretations have been developed.

Definitions (SOLAS II-2/3)

Clear categorization was introduced:

- Pantries without cooking appliances were defined as accommodation spaces; and
- Pantries with cooking appliances were defined as service spaces.

Explicit power thresholds and examples were included.

Fire containment philosophy

The purpose of the regulation was retained:

- to contain fire within the space of origin; and
- to maintain fire integrity at boundaries, openings and penetrations.

Stairway arrangements

- A clear distinction was introduced between:
 - stairways penetrating a single deck;
 - stairways penetrating multiple decks (addressed in a new sub-paragraph).
- Terminology was standardized:
 - “stair” was replaced with “stairway”;
 - “superimposed” was replaced with “stairway above”; and
- Supporting figures were to be developed in an MSC circular.

Dumbwaiters

- Dumbwaiters were treated as lifts rather than stairways; and
- Requirements were placed in dedicated paragraphs across ship types.

This reinforced the mandatory structural fire integrity requirements.

Structural fire integrity tables (Tables 9.1–9.8)

- The notes were clarified to be binding regulatory provisions;
- They were to be applied together with the prescribed fire integrity ratings; and
- Their formatting was to be revised to avoid confusion with footnotes.

Watertight doors in fire divisions

- The wording was revised while retaining references to SOLAS II-1/13 and 16;
- The exemption for doors below the bulkhead deck was maintained; and
- Doors above the bulkhead deck were required to meet the fire integrity of the division.

Ventilation ducts

8.1 Material “equivalent to steel”

- Criteria were introduced for non-steel ducts to be considered equivalent; and

- These provisions were incorporated into SOLAS to support objective verification during design approval.

8.2 Ducts penetrating “A” class divisions

- A requirement was introduced that the duct section between the division and the fire damper be insulated to the same fire integrity as the penetrated division;
- Concerns were expressed regarding:
 - additional insulation requirements;
 - potential industry burden;
- The amendment was nevertheless retained.

Ventilation rooms (machinery spaces category A)

- Proposed amendments were not retained; and
- It was considered that existing SOLAS provisions already addressed the matter.

Overall regulatory nature

The amendments:

- incorporated Unified Interpretations into SOLAS;
- clarified previously ambiguous provisions; and
- introduced certain additional technical requirements.

Outstanding issues

The application of the draft amendments was difficult to determine because the revision combined different elements, including the incorporation of existing Unified Interpretations, clarifications of previously ambiguous provisions, and the introduction of certain new technical requirements, making it challenging to distinguish which parts should apply to new ships only and which could be applied to existing ships. As a result, the Working Group did not finalize the application provisions and agreed that this issue should be further developed intersessionally by the Correspondence Group.

Further work

FP Correspondence Group 1 will further develop the amendments, particularly the application provisions.

Validation of model courses

SSE 12 validated the Model Course 3.06 on Survey of Life-Saving Appliances and Arrangements.

Detection and control of fire on container ships

The SSE Sub-Committee has been working on several measures to address fires onboard container ships.

Detection

SSE 12 discussed the potential use of advanced detection technologies, including video-based fire detection systems and other emerging methods capable of identifying fire or smoke at an early stage. The discussion recognized that further technical development and evaluation would be required before such technologies could be incorporated into the regulatory framework. SSE 12 therefore supported continuing technical work on detection technologies in parallel with the development of suppression measures.

Nevertheless, SSE 12 progressed draft amendments to SOLAS regulation II-2/7.11, introducing requirements for portable infrared thermal imagers. These devices are intended to support the detection, confirmation and localisation of fires, particularly for containers carried on or above the weather deck, by enabling the identification of hot spots and affected areas that may not be visible through conventional means. The amendments address aspects such as the number, storage, testing and functionality of the equipment, with reference to relevant standards. The draft provisions were developed to enhance the effectiveness of onboard fire response by improving situational awareness, rather than replacing existing fire detection systems. MSC 112 will review the text for approval and subsequent adoption.

Suppression

SSE 12 further considered suppression measures intended to improve the capability to control container fires both on deck and within cargo holds.

Fixed and mobile water monitors

SSE 12 considered the use of water monitor systems to enable the application of water to container stacks from a safe distance as part of the package of firefighting devices intended to enhance manual firefighting capability for deck container fires.

Due to time constraints, SSE 12 could not come to a concrete agreement.

Water mist lances

SSE 12 agreed draft amendments to SOLAS regulation II-2/10.7.3.1 on water mist lances, specifying that water mist lances to be provided on board ships constructed on or after 1 January 2032 shall be of the self-penetrating type. The amendments also include transitional flexibility, allowing ships constructed on or after 1 January 2016 and before 1 January 2032 to apply the more recent requirements. MSC 112 (December 2026) will review the SOLAS amendments for approval and subsequent adoption. SSE 12 also developed draft guidelines for the design, performance, testing and approval of water mist lances for approval in conjunction with the SOLAS amendments. SSE 12 also agreed that the Sub-Committee on Implementation of IMO Instruments (III) should consider verification aspects related to these requirements.

Fire hydrant water distributors

SSE 12 noted proposals concerning fire hydrant water distributors intended to allow the output of a single hydrant to be divided into multiple hose lines. Further consideration of this matter was referred to the FP Correspondence Group 2.

Active protection systems below hatch covers

SSE 12 noted proposals concerning the introduction of active protection systems for container cargo holds below hatch covers, including the possible use of water spray systems. Further technical consideration of this matter was referred to FP Correspondence Group 2.

Other issues

SSE 12 agreed that, once the draft SOLAS amendments on active protection systems below hatch covers are finalized, input should be sought from the Sub-Committee on Carriage of Cargoes and Containers (CCC) on whether the introduction of water spray systems would require transport provisions for water-sensitive cargoes, and from the Sub-Committee on Ship Design and Construction (SDC) on whether modifications to the bilge system might be necessary.

PFOS and PFAS Ban

SSE 12 recalled that MSC 107 had adopted amendments to SOLAS and the 1994 and 2000 HSC Codes prohibiting the use and storage of fire-extinguishing media containing perfluorooctane sulfonic acid (PFOS) and had expanded the scope of the related output to consider prohibiting other fluorinated fire-fighting foams.

Proposal to prohibit PFAS in SOLAS chapter II-2 and the 1994 and 2000 HSC Codes

SSE 12 discussed the prohibition of fire-fighting foams containing per- and polyfluoroalkyl substances (PFAS) in SOLAS chapter II-2 and the 1994 and 2000 HSC Codes. PFAS refers to a large family of fluorinated chemicals, estimated to include around 12,000 substances, of which PFOS (perfluorooctane sulfonic acid) is one specific compound already prohibited under SOLAS. While several delegations supported moving towards a prohibition due to environmental and health concerns, differing views were expressed on whether the ban should apply to the entire PFAS class or follow a substance-specific approach consistent with the Stockholm Convention. Concerns were also raised regarding the effectiveness of fluorine-free alternatives, transition periods, detection methods and implementation impacts. SSE 12 concluded that further work was required and agreed to continue consideration of the issue at SSE 13.

SSE 12 tasked the CG to further consider this issue.

Maintenance, thorough examination, operational testing, overhaul and repair of lifeboats and rescue boats, launching appliances and release gear (resolution MSC.402(96))

Triggered by discussions at the Maritime Safety Committee (MSC) on the definitions of “make” and “type”, the SSE Sub-Committee continued its comprehensive review of resolution MSC.402(96), including the associated definitions. While ISO PAS 23673:2020 provided details of “make” and “type”, it was noted that ISO 23678:2022 deviated from these definitions. SSE 12 therefore agreed, for the time being, to delete the footnote reference to ISO 23678:2022, noting that the standard may be referenced again once it is updated to align with the revised definitions in the LSA Code.

Definition of make, type, series and model

SSE 11 (March 2025) developed definitions of Make, Type, Series and Model based on the following understanding, and agreed in principle, for further consideration:

EXAMPLES OF "MAKE", "TYPE", "MODEL" AND "SERIES"
(emanating from document SSE 9/19/8)

Make	Type	Series	Model
Launching appliances (incl. davits, ramps)			
FASSMER	Hydraulic davit	FHD	FHD 150, FHD 280, FHD 450
FASSMER	Hinged davit	FHDY	FHDY 100
FASSMER	Freefall davit	FFH	FFH 65, FFH 70, FFH 80, FFH 90

- Make – original manufacturer of the type, model and series of equipment, as referred to on the approval certificate and/or ID plate, as appropriate;
- Type – category of equipment having common functional or design;
- Model – a specific version of a particular make and type, as referred to on the approval certificate and/or ID plate, as appropriate; and
- Series – a specific range of models from the same manufacturer that have equivalent design characteristics and maintenance requirements.

Training and qualification

SSE 12 considered draft amendments to paragraph 7.1.1 of resolution MSC.402(96) concerning the certification and documentation of personnel performing maintenance, examination, testing and repair of life-saving appliances. During the discussion, it was agreed to delete the term “recognized” and not to include the qualifier “as appropriate”, due to concerns regarding ambiguity and inconsistent implementation. Differing views were expressed regarding the phrase “who inferiors to those acceptable to the Organization”, which was retained in square brackets pending further clarification. Due to time constraints, the Sub-Committee did not finalize this matter and agreed to continue the work by correspondence.

Thermal performance of immersion suits

There was an information paper proposed incorporating requirements for evaluating the thermal performance of immersion suits using thermal manikins. SSE 12 noted progress in the revision of the ISO 15027 series. Given that the ISO standard was still under development, SSE 12 considered it premature to complete the work and therefore addressed only limited elements at this session, agreeing in principle to the use of clo values (clothing insulation value) as an acceptance criterion for immersion suits and deciding not to amend the LSA Code under this output, while further technical discussion, including the pencil pick-up test and possible amendments to resolution MSC.81(70) and MSC.1/Circ.1628/Rev.4, was deferred to SSE 13.

Carriage of new energy vehicles in special category and ro-ro spaces

This agenda item addresses ship-type-specific matters and general safety issues of the carriage of new energy vehicles. Carriage of battery electric vehicles (BEVs) has also been addressed by the Sub-Committee on Carriage of Cargoes and Containers (CCC) in conjunction with the IMDG Code.

Scope

SSE 12 agreed that the scope of the work on new requirements related to the carriage of new energy vehicles should remain unchanged and should cover all relevant categories of ro-ro ships. SSE 12 agreed that any measures to be developed should reflect the fundamental differences between ro-ro cargo ships and ro-ro passenger ships while ensuring consistency with the amendments adopted by resolution MSC.550(108). SSE 12 further agreed that the work should address three categories of ships in order of priority, namely vehicle carriers (including pure car and truck carriers), other ro-ro cargo ships, and ro-ro passenger ships, with the work to focus first on cargo ships before considering passenger ships as appropriate.

Fire detection

The CG established by SSE 11 (2025) was of the view that early warning and/or detection of fires would be instrumental in improving the fire safety of ships carrying new energy vehicles. SSE 12 further addressed that matter.

Detection technologies

SSE 12 considered proposals related to video-based monitoring systems and integrated infrared and image-based fire detection concepts for use in vehicle and ro-ro spaces. Information was presented indicating that such technologies could support early detection, confirmation and localisation of fires, including through the use of image processing techniques. While there was support in principle for further development, concerns were expressed regarding technical limitations, such as delayed detection, obscured camera images and blind spots. SSE 12 noted that such proposals could contribute to the development of future base texts; however, no draft regulatory provisions were developed at this stage, and the work remains at a preliminary stage requiring further consideration.

Fire suppression

Various proposals were submitted addressing fire suppression arrangements for ships carrying new energy vehicles, including the performance of existing fixed fire-fighting systems and potential alternative approaches. However, no agreement was reached at this stage on specific technical solutions or on the selection of suppression technologies.

The discussions highlighted the challenges associated with fires involving new energy vehicles, in particular the difficulty in controlling such fires once they have developed, including the potential for re-ignition and fire spread.

Action plan

SSE 12 agreed on a revised action plan setting out a structured approach for the development of measures addressing the fire safety of ships carrying new energy vehicles. The plan focuses on the development of goal-based and functional requirements covering fire prevention, early detection, fire containment and fire suppression, including the capability to control and cool affected and adjacent vehicles and to mitigate the risk of fire spread and re-ignition. As an initial priority, SSE 12 agreed to develop comprehensive interim fire safety guidelines for pure car and truck carriers (PCTCs), with the development of draft SOLAS amendments to follow at a later stage.

Safety regulatory framework for alternative fuels and new technologies for GHG emission reduction (Agenda item 17)

SSE 12 initiated work on the safety regulatory framework for alternative fuels and new technologies following instructions from MSC 110 to address regulatory gaps and barriers associated with emerging propulsion and energy systems.

Work plan for the development of the GHG Safety regulatory framework

SSE 12 developed and agreed on a draft work plan for the development of a safety regulatory framework to support the reduction of GHG emissions from ships using new technologies and alternative fuels under the purview of the Sub-Committee. In preparing the work plan, the Working Group noted the need for appropriate working arrangements at future SSE sessions, highlighted that safety work on fuels such as Fischer-Tropsch (FT) diesel and hydrotreated vegetable oil (HVO) should consider their different chemical characteristics and the related work of the CCC Sub-Committee, and expressed views that SOLAS provisions may not always be the most appropriate regulatory vehicle for fuel-specific safety measures. The agreed work plan sets out a multi-year programme under the SSE Sub-Committee, including development of standards for fire-fighting systems capable of extinguishing alcohol fires under the FSS Code, development of SOLAS chapter II-2 amendments addressing the fire risks associated with FT diesel fuels, and the development of interim safety guidelines for advanced waste heat recovery systems, with endorsement and approval milestones scheduled through MSC sessions up to 2030.

Approach and structure for interim guidelines on battery energy storage systems

SSE 12 considered proposals on the approach, scope and structure of interim guidelines for ships using battery energy storage systems (BESS). SSE 12 agreed that the guidelines should be technology-neutral and goal-based. Discussions highlighted the need to carefully define battery-related terminology, consider different battery chemistries and thresholds such as the existing 20 kWh compartment capacity benchmark, and address training and hazard-based safety considerations. SSE 12 established the correspondence group, to develop draft technology-neutral interim guidelines for battery energy storage systems and prepare definitions for relevant battery technologies, with a report to be submitted to SDC 13.

Further information

For further information please contact: imo@lisr.com

Annex

Provisional list of draft circular/resolution

- Draft amendments to MSC.1/Circ.1318/Rev.1 on Revised guidelines for the maintenance and inspection of fixed carbon dioxide fire-extinguishing systems;
- Draft amendment to paragraph 2.1.2.5 of chapter 5 of the FSS Code
- Draft amendments to the 2010 Fire Test Procedures Code (2010 FTP Code) to introduce fire test procedures for plastic pipes in annex 1, part 6
- Draft amendments to SOLAS regulation II-2/7.11 on infrared thermal imagers
- Draft amendments to SOLAS regulation II-2/10.7.3.1 on water mist lances
- Draft guidelines for the design, performance, testing and approval of water mist lances
- Draft amendments to MSC.1/Circ.1628/Rev.4 on Revised standardized life-saving appliance evaluation and test report forms (personal life-saving appliances)
- Draft amendments to MSC.1/Circ.1629 on Revised standardized life-saving appliance evaluation and test report forms (visual signals)
- Draft amendments to MSC.1/Circ.1630/Rev.3 on Revised standardized life-saving appliance evaluation and test report forms (survival craft)
- Draft amendments to MSC.1/Circ.1633 on Revised standardized life-saving appliance evaluation and test report forms (other life-saving appliances)
- Model Course 3.06 on Survey of Live saving appliance